

Unit 1 — Foundations & Intelligent Agents

Review deck for the test on Thu Oct 8

English for the exam · "ES:" tips in Spanish

Wiki: What is AI · History of AI · Intelligent Agents · Task Environments · Agent Types · State Representation

What is AI?

- No single definition of *intelligence*; the course lists **understanding, problem solving, knowledge, meaning, skill**.
- **Turing Test (1950)**: an operational, behavior-based definition — can a machine pass as human in conversation?
- **Dartmouth (1956)**: "every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it."
- **AIMA's view**: AI = building **rational agents**.

ES: si preguntan "¿qué es IA?", responde con *rational agent* + menciona Turing y Dartmouth.

Timeline you must know

Year	Who	What
1950	Alan Turing	Turing Test
1956	McCarthy, Minsky, Rochester, Shannon	Dartmouth workshop, "AI" named
1958	Frank Rosenblatt	Perceptron
1972	Colmerauer & Roussel (Kowalski's theory)	Prolog , Marseille
1975	John Holland	Genetic algorithms
1995	Kennedy & Eberhart	Particle swarm optimization
1996	Dorigo, Maniezzo & Coloni	Ant System (ACO)
2007	Karaboga	Artificial bee colony

ES: trampas de las slides → Prolog **no** es de Dennis Ritchie (eso es C); GA = **1975**, no 1992.

Agent, agent function, agent program

- **Agent:** perceives the environment through **sensors**, acts through **actuators**.
- **Agent function:** maps the whole *percept history* → action (abstract, possibly infinite table).
- **Agent program:** the concrete code that implements the function compactly.
- **Rational agent:** chooses the action that maximizes the **expected** value of the performance measure, given the percepts so far and its built-in knowledge.

ES: racional $\neq$ omnisciente. Maximiza el desempeño **esperado**, no el real. Cruzar la calle tras mirar es racional aunque luego caiga una puerta de avión.

Performance measures: "you get what you ask for"

- A vacuum robot rewarded **+1 per unit of dirt vacuumed** could dump dirt and vacuum it again.
- Better: **+1 per clean square per time step** (minus energy/noise penalties).

Say it in the exam: "Design performance measures according to what one actually wants in the environment, not according to how one thinks the agent should behave."

PEAS — specify the task environment first

	Performance	Environment	Actuators	Sensors
Sudoku app	correct board, time	9×9 board with clues	write digits, show solution	file / camera
Robot soccer	goals for – against	field, ball, teammates, rivals	legs/wheels, kicker	camera, odometry, touch
Shopping assistant	price, quality, user satisfaction	web shops, products, user	show products, buy, ask	web pages, user input

ES: PEAS es siempre el primer paso para diseñar un agente.

Six environment dimensions

1. **Fully** vs. **partially** observable
2. **Deterministic** vs. **non-deterministic (stochastic)**
3. **Episodic** vs. **sequential**
4. **Static** vs. **dynamic** (*semidynamic*: the world doesn't change but the score does with time)
5. **Discrete** vs. **continuous**
6. **Single-agent** vs. **multi-agent** (competitive / cooperative)

Classical search assumes: *fully observable, deterministic, static, discrete, known.*

Classify these (practice U1 #2)

Task	Obs.	Det.	Episodic	Static	Discrete	Agents
Crossword	full	det.	seq.	static	discrete	single
Poker	partial	stoch.	seq.	static	discrete	multi
4×4 tic-tac-toe	full	det.	seq.	static	discrete	multi (comp.)
Image classification	full	det.	episodic	semi	continuous	single
Delivery drone	partial	stoch.	seq.	dynamic	continuous	multi

ES: clasificar imágenes es **episódico**: cada imagen no depende de la anterior.

Four agent architectures (+ learning)

Type	Uses	Example
Simple reflex	condition-action rules on the <i>current</i> percept	thermostat
Model-based reflex	internal state + transition model	vacuum that maps the house
Goal-based	search / planning toward a goal	GPS route planner
Utility-based	maximizes expected utility; trades off conflicting goals	self-driving taxi

Learning agent: any of the above + learning element, critic, problem generator (e.g., a chess program that improves by self-play).

Is the vacuum agent rational? (AIMA §2.2)

- Two squares; rule: *if dirty → Suck, else move to the other square.*
- Measure **+1 per clean square per step** → **rational** (no agent does better in expectation).
- Add **−1 per move** → **not rational**: it keeps oscillating when everything is clean.
- Fix: a **model-based** agent that remembers both squares are clean and does *NoOp*.

State representations

Representation	What a state is	Course example
Atomic	an indivisible black box	"Sibiu" in Romania route finding
Factored	a vector of variables with values	8-puzzle tuple, Sudoku as a CSP, (x, y) in optimization
Structured	objects and relations	<code>parent(hector, ana).</code> in Prolog / FOL

More expressive → more reasoning possible → more expensive.

Statistical vs. causal models

- **Model:** a mathematical description of a system / hypothesis.
- **Statistical** (regression, classification, Bayesian/Markov networks): capture **correlation**.
- **Causal** (Judea Pearl's causal calculus): cause $\rightarrow$ effect in a **DAG**; presented as the bridge between ML and AI.

ES: "correlación no implica causalidad" — el modelo causal permite preguntar "¿qué pasa si intervengo?".

Self-check (answer aloud, in English)

1. Define a rational agent in one sentence.
2. Why is a rational agent not always successful?
3. PEAS for an automated taxi.
4. Which agent type is a GPS app, and why?
5. Is poker episodic? Is it fully observable?
6. Who created Prolog, and when?

Answers: wiki [study/practice-unit-1-agents.md](#) · more drills in the Exam Drill quiz